

Quiz (Carolina Reaper)

- Q1.** A shipping company has 3 : 5 ratio of trucks to vans. If they have 18 vans, how many trucks are there?
(A) 6
(B) 8
(C) 10
(D) 12
(E) 15
- Q2.** The traffic flow ratio of cars to buses is 2 : 7. If there are 14 buses, how many cars are there on the road?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q3.** A logistics company has a ratio of x : 15 of delivery trucks to warehouses. If $x = 5$, how many warehouses are needed for 25 delivery trucks?
(A) 60
(B) 75
(C) 50
(D) 30
(E) 25
- Q4.** The ratio of a : b of cargo ships to speedboats is 8 : 3. If the number of speedboats is 9, how many cargo ships are there?
(A) 16
(B) 20
(C) 24
(D) 28
(E) 32
- Q5.** If 12 boxes of cargo weigh 3 : 4 ratio of kilograms to pounds, and x kilograms weigh y pounds, what is the ratio of x : y ?
(A) 3 : 4
(B) 4 : 3
(C) 12 : 16
(D) 16 : 12
(E) 9 : 12
- Q6.** A transportation company has a ratio of 7 : 3 of diesel to gasoline trucks. If there are 24 diesel trucks, how many gasoline trucks are there?
(A) 6
(B) 8
(C) 9
(D) 10
(E) 12
- Q7.** A ratio of 9 : 5 represents the number of wheels on cars to wheels on motorcycles. If x is the number of wheels on 35 motorcycles, how many wheels are on the cars?
(A) 45
(B) 54
(C) 63
(D) 72
(E) 81
- Q8.** If a logistics company has 4 : 9 ratio of large to small packages and they have 36 small packages, how many large packages are there?
(A) 12
(B) 16
(C) 20
(D) 24
(E) 28

Open Ended Questions (FRQ)

- Q9.** A truck is traveling from City A to City B at a constant 60 mph. If the distance between the two cities is 210 miles, and the truck has 3 stops for fuel with a 1 : 5 ratio of fuel to distance traveled, how much fuel does the truck use in total for the entire trip? Show your work.
- Q10.** A boat's cargo is in a 2 : 3 ratio of crates to barrels. If the boat has 24 crates, what is the total weight of cargo if each crate weighs 50 pounds, and each barrel weighs 75 pounds? Show all calculations.

Chef's Tips

- Tip 1: Make sure to check units for all calculations.
- Tip 2: Remind students to label all parts of their work.
- Tip 3: Allow students to use calculators only for the last question.